

SECTION C

11 (**long**) The expected excess sunspot count over a solar cycle of duration T is given by the function

$$f(t) = \begin{cases} A \left(1 - \cos \left(\frac{\pi t}{\alpha T} \right) \right) & \text{for } 0 \leq t \leq \alpha T \\ A \left(1 + \cos \left(\frac{\pi(t - \alpha T)}{T - \alpha T} \right) \right) & \text{for } \alpha T \leq t \leq T \end{cases}$$

where $0 < \alpha < 1$ indicates the relative position of the solar peak within a cycle.

(a) Make a rough sketch of the function for $\alpha = 1/3$. [5]

(b) Specify the constant term d of the Fourier series expansion of $f(t)$ and justify your answer based on the sketch. [5]

(c) What is the Fourier series expansion of $f(t)$ when $\alpha = \frac{1}{2}$? [5]

(d) For $\alpha \neq \frac{1}{2}$, it can be shown that the Fourier series coefficients decay as $O(n^{-3})$. Explain why this is the case. [8]

(e) The following two periodic functions are identical to $f(t)$ for $0 \leq t \leq T$

$$f_1(t) = A + \frac{A}{\pi} \sum_{n=1}^{\infty} u_n \sin(\pi n \alpha) \cos \left(\frac{\pi n t}{T} \right)$$

$$f_2(t) = \frac{A}{\pi} \sum_{n=1}^{\infty} (v_n + w_n \cos(\pi n \alpha)) \sin \left(\frac{\pi n t}{T} \right)$$

for suitably defined u_n , v_n and w_n . Make a rough sketch of $f_1(t)$ and of $f_2(t)$ over one of their respective periods. [7]